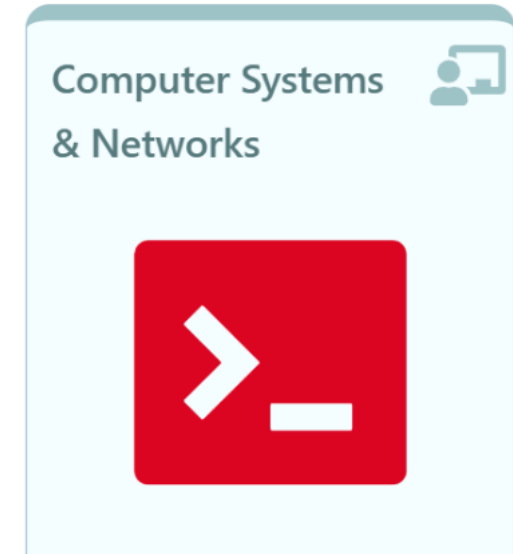
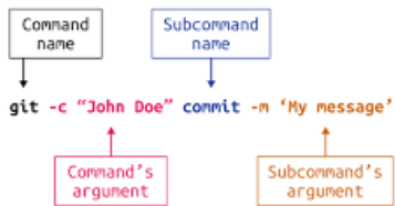


Computer Systems & Networks



01: Command Line



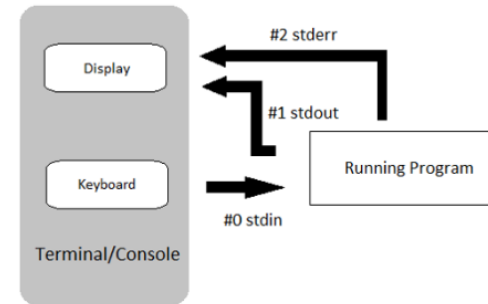
Commands · Arguments ·
Options · Variables

02: File Management



Globs · Copy · Move ·
Remove

03: IO Redirection



STDIN · STDOUT · STDERR

Lab



File Management · IO
Redirection · Search &
Filter text

02: File and Text Manipulation



Redirection · grep ·
Positional Param · Exit
Codes

Working with Files

Learning Outcomes:

- This class will cover the basics of the file management such as:
 1. Globbing
 2. Copying files
 3. Moving files
 4. Renaming files
 5. Removing files

Compliment
your learning
with Chapter 8

Linux Essentials

Content

Objectives

Key Terms

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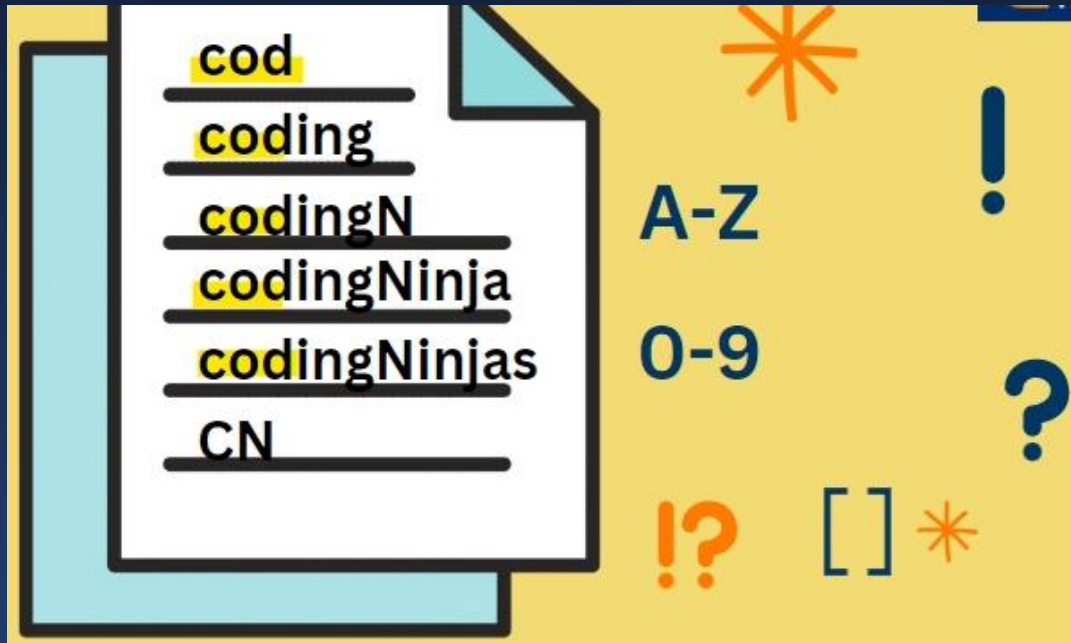
8.1 Introduction

When working in a Linux Operating System, it is important to know how to manipulate files and directories. Some Linux distributions have GUI-based applications that allow you to manage files, but it is advantageous to know how to perform these operations via the command line.

Note

It is important to note that everything in Linux is case-sensitive, a feature carried over from Unix. This means that the shell recognizes a lowercase `a-z` character as completely different from an uppercase `A-Z` character. When manipulating files, pay attention to your capitalization: a `hello.txt` file is different from `HELLO.txt` and `Hello.txt` files.

The character standard used for Linux is the UTF-8 standard which is based on the ASCII (American Standard Code for Information Interchange). To learn more about ASCII, execute the `man -s 7 ascii` command.



1. Glob characters



Glob characters

Glob characters are often referred to as *wild cards*.



These are **symbol characters** that have special meaning to the shell.



Globs are powerful because they allow you to specify patterns that match filenames in a directory:

= Instead of manipulating a single file at a time, you can easily execute commands that will affect many files.

*Glob character: The Asterisk **

- The asterisk character is used to represent zero or more of any character in a filename.

```
compsys@compsys-virtualbox:~$ ls -l
total 52
drwxrwxr-x 2 compsys compsys 4096 Aug 31 2022 Desktop
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Documents
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Downloads
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Music
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Pictures
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Public
drwxrwxr-x 2 compsys compsys 4096 Aug 30 12:33 Sept
drwx----- 3 compsys compsys 4096 Aug 25 21:34 snap
-rwxr-x--x 1 compsys compsys 248 Aug 29 13:54 stats
-rwxr-xr-x 1 compsys compsys 29 Aug 31 10:55 tempdate
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Templates
drwxrwxr-x 3 compsys compsys 4096 Aug 31 13:01 tmp
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Videos
compsys@compsys-virtualbox:~$ touch Semester1 Semester1Results Semester2 Semester3 SemanticDetails
compsys@compsys-virtualbox:~$ ls -l Sem*
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 SemanticDetails
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1Results
```


Glob character: The Question Mark ?

The question mark character matches **exactly one character**, no more and no less.

```
compsys@compsys-virtualbox:~$ ls -l Sem?????  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester2  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester3  
compsys@compsys-virtualbox:~$
```

The asterisk and question mark could also be used together to look for files

- For example: to look for files with three-letter extensions in the etc directory , use

```
echo /etc/*.???
```

Glob character: **Brackets []**

used to match a single character by representing a range of characters that are possible match characters.

```
compsys@compsys-virtualbox:~$ ls [SD]*
SemanticDetails  Semester1  Semester1Results  Semester2  Semester3

Desktop:
computer.desktop  network.desktop  qterminal.desktop  trash-can.desktop  user-home.desktop

Documents:

Downloads:

Sept:
compsys@compsys-virtualbox:~$
```

Brackets can also be used to represent a range of characters by using the - character

```
compsys@compsys-virtualbox:~$ ls [S-W]*
```

Another common use is matching numbers i.e. using the [0-9] pattern

Using ls command with Glob characters

- Use the `-d` option, which tells the `ls` command to display the name of directories, instead of their contents:

```
compsys@compsys-virtualbox:~$ ls -d [SD]*
Desktop Documents Downloads SemanticDetails Semester1 Semester1Results Semester2 Semester3 Sept
compsys@compsys-virtualbox:~$
```

Glob character: **Exclamation Point !**

- The exclamation point is used in conjunction with the square brackets to negate a range.

```
compsys@compsys-virtualbox:~$ ls [!S]*
stats  tempdate

Desktop:
computer.desktop  network.desktop  qterminal.desktop  trash-can.desktop  user-home.desktop

Documents:

Downloads:

Music:

Pictures:

Public:

snap:
firefox

Templates:

tmp:
tutorial

Videos:
compsys@compsys-virtualbox:~$
```

Copying Files

cp



Copying Files

- The `cp` command is used to copy files. It requires a source and a destination.

```
cp [source] [destination]
```



`cp` will overwrite any existing files by default



```
compsys@compsys-virtualbox:~$ cp Semester1Resu* Sept
compsys@compsys-virtualbox:~$ ls Sept
Semester1Results
compsys@compsys-virtualbox:~$ █
```

Copying Files – v (Verbose Mode)

- The `-v` option will cause the `cp` command to produce output if successful.
- An example of the `-v` option used with the `cp` command:
- When the destination is a directory, the resulting new file will have the same name as the original file.
 - If you want the new file to have a different name, you must provide the new name as part of the destination.

```
compsys@compsys-virtualbox:~$ cp -v Semester2 Sept  
'Semester2' -> 'Sept/Semester2'
```

Copying Files - Avoid Overwriting Data

- The `cp` command can be destructive to existing data if the destination file already exists.

```
compsys@compsys-virtualbox:~$ cp -vi Semester2 Sept  
cp: overwrite 'Sept/Semester2'? █
```

- With the `-i` (interactive) option, the `cp` will prompt before overwriting a file (`y` (yes) or `n` (no))

Copying Directories

- Using the **cp** command to copy directories will result in an error message:

```
sysadmin@localhost:~$ cp -n /etc/skel/. * ~  
cp: omitting directory `/etc/skel/.'  
cp: omitting directory `/etc/skel/..'
```

- However, the **-r** (recursive) option to the **cp** command will have it copy both files and directories.



Be careful

The entire directory structure will be copied.

This could result in copying a lot of files and directories!



Moving/ Renaming Files

mv



Moving Files: the **mv** command

- The syntax for the **mv** command is much like the **cp** command:

```
mv [source] [destination]
```

- The **mv** command also renames a file.
 - The name of the file will change only if a destination file name is also specified.

```
compsys@compsys-virtualbox:~$ mv Semester1 Sept/ClassList
compsys@compsys-virtualbox:~$ ls Sept
ClassList Semester1Results Semester2
compsys@compsys-virtualbox:~$
```

Moving Files - Additional **mv** Options

- Like the **cp** command, the **mv** command provides the following options:

Option	Meaning
-i	Interactive: Ask if a file is to be overwritten.
-n	Do not overwrite a destination files' contents.
-v	Verbose: Show the resulting move.

Removing Files



Deleting Files

To delete a file, use the `rm` command



Be careful if using glob/wildcard characters



```
compsys@compsys-virtualbox:~$ ls
Desktop  Downloads  Pictures  SemanticDetails  Semester2  Sept  stats  Templates  Videos
Documents Music      Public    Semester1Results Semester3  snap  tempdate tmp
compsys@compsys-virtualbox:~$ rm Sem*
compsys@compsys-virtualbox:~$ ls
Desktop  Downloads  Pictures  Sept  stats  Templates  Videos
Documents Music      Public    snap  tempdate tmp
compsys@compsys-virtualbox:~$
```

Deleting Directories

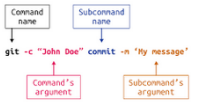
- The `rm` command can be used to delete directories. However, the default usage (no options) of the `rm` command will fail to delete a directory:

```
compsys@compsys-virtualbox:~$ ls tmp
tutorial
compsys@compsys-virtualbox:~$ rm -i tmp
rm: cannot remove 'tmp': Is a directory
compsys@compsys-virtualbox:~$ rm -ri tmp
rm: descend into directory 'tmp'? y
rm: descend into directory 'tmp/tutorial'? y
rm: remove directory 'tmp/tutorial/Oct'? y
rm: remove directory 'tmp/tutorial/Nov'? n
rm: descend into directory 'tmp/tutorial/Dec'? y
rm: descend into directory 'tmp/tutorial/Dec/Part1'? y
rm: descend into directory 'tmp/tutorial/Dec/Part1/Outputs'? y
rm: remove regular empty file 'tmp/tutorial/Dec/Part1/Outputs/Part1.txt'? y
rm: remove regular file 'tmp/tutorial/Dec/Part1/Outputs/output.txt'? █
```

- To delete a directory, use the `-r` (recursive) option to the `rm` command:
Important: recall the `-i` option, why?

Try out this weeks Lab

01: Command Line



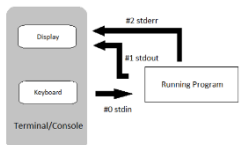
Commands · Arguments · Options · Variables

02: File Management



Globs · Copy · Move · Remove

03: IO Redirection



STDIN · STDOUT · STDERR

Lab



File Management · IO Redirection · Search & Filter text

Lab

Computer Systems & Networks

Current Week 1

Search

Computer Systems & Networks

02: File and T...

01: File Manip...

Lab

02. Copy Move Remove

Copy · Move · Remove

Copy Using `cp` command

- Have a look inside the `/etc/` directory
- Copy this `/etc/hosts` file to your home directory

```
cp /etc/hosts ~ # Recall the ~ sign
```

- check it's copied HOW?? ANS: Use the `___` command

Verbose Mode `cp -v`

Use the `-v` flag to confirm the success of the actions:

```
cp -v /etc/hosts ~/tmp
```

- If you don't use `-v`, the `cp` command will copy the files silently (i.e., no output if successful).

EXERCISES using `cp`

Make another copy inside your `~/tmp` directory, returning the success of the action: